

MEESHO RESELLER

REVENUE ANALYTICS

Reliable Data Analysis, Growth Monitoring
and AI-Assisted Business Narrative

Project Type	Data Analytics / Reliable AI Analytics
Analysis Period	April 2026 – June 2026
Technology	Python • Pandas • SQLite • SQL • VS Code
API Requirement	No external API key required

Capstone Project Report

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1. Executive Summary

The Meesho Reseller Revenue Analytics project demonstrates an end-to-end data analytics workflow for analyzing reseller order and revenue information. The project transforms raw reseller order data into verified business insights through SQL analysis, Month-on-Month growth monitoring, reliable narrative generation, data masking, and a local revenue analysis agent.

The project uses a simulated dataset containing 900 orders covering April, May, and June 2026. The dataset includes 24 resellers distributed across four regions, with three cities represented in each region.

- Beauty & Personal Care grew by 13.64% from April to May.
- Ethnic Wear grew by 17.00% from April to May.
- Western Wear grew by 9.63% from April to May.
- Ethnic Wear grew by 35.48% from May to June.

The project includes validation of source data, growth calculations, reseller identifier masking, negative testing, and safe failure when required input files are unavailable. The complete workflow operates locally and does not require an external API key.

2. Business Problem

Reseller businesses generate large volumes of transactional information through orders, products, quantities, prices, revenue, returns, cancellations, and reseller activity. Raw transactional data does not by itself provide management with immediate answers about category growth, regional revenue, reseller contribution, or safe communication of analytical results.

This project builds a structured workflow that converts transactional data into reliable and understandable business information while keeping reported insights connected to verified source data.

3. Project Objectives

- Generate a structured reseller order dataset.
- Store and analyze the data using SQLite.
- Calculate category-wise and region-wise revenue.
- Calculate Month-on-Month revenue growth.
- Identify categories exceeding the defined 8% growth threshold.
- Generate concise evidence-based business narratives.
- Mask reseller identifiers before narrative processing.
- Build a local revenue analysis agent.
- Validate calculations and outputs.
- Demonstrate safe failure when required inputs are missing.
- Maintain a complete and reproducible project workflow.

4. Dataset Description

The project uses a simulated reseller order dataset.

Item	Value
Total Orders	900
Analysis Period	April–June 2026
Orders per Month	300
Total Resellers	24
Regions	4
Cities per Region	3
Product Categories	5
Duplicate Order IDs	0

Product categories: Ethnic Wear; Western Wear; Kids Wear; Home & Kitchen; Beauty & Personal Care.

Order status distribution: Delivered 70%; Returned 15%; Cancelled 10%; Pending 5%.

A fixed random seed is used for deterministic and reproducible dataset generation.

5. Technology Stack

Technology	Purpose
Python	Data generation and analytical processing
Pandas	Data handling and CSV processing
SQLite	Local database storage and SQL analysis
SQL	Revenue and business analysis
VS Code	Development environment
Markdown	Project documentation
CSV	Analytical data exchange

6. Project Architecture

The project follows a sequential analytical pipeline:

Stage	Output
Dataset Generation	Structured reseller order data
Part 1 — SQL	Verified revenue and order metrics
Part 2 — Growth Engine	MoM growth and threshold flags
Part 3 — Narrative	Masked data and evidence-based narrative
Part 4 — Agent	Validated business narrative
Final Reporting	Management-ready documentation

7. Part 1 — SQL Analysis

Part 1 establishes the verified analytical foundation. The orders table contains order_id, order_date, reseller_id, reseller_name, region, city, category, quantity, unit_price, order_amount, order_status, and revenue.

Reports include monthly category revenue, region-wise revenue and order count, top five resellers by total spend, June delivered-order AOV, and a duplicate Order ID check.

Verified data quality result: Duplicate Order IDs = 0.

8. Part 2 — Growth Engine

Month-on-Month growth is calculated as:

$$\text{MoM Growth \%} = ((\text{Current Revenue} - \text{Previous Revenue}) / \text{Previous Revenue}) \times 100$$

The project applies an 8% threshold: growth greater than 8% is flagged; exactly 8% is assigned an escalation boundary status; values below 8% are not flagged. Negative growth is not automatically flagged under this rule.

May 2026 compared with April 2026

Category	April Revenue	May Revenue	Growth	Flag
Beauty & Personal Care	207,421.46	235,710.83	13.64%	Flagged
Ethnic Wear	421,858.43	493,572.34	17.00%	Flagged
Home & Kitchen	619,987.73	494,285.21	-20.28%	Not Flagged
Kids Wear	241,899.28	243,977.89	0.86%	Not Flagged
Western Wear	391,501.83	429,203.55	9.63%	Flagged

June 2026 compared with May 2026

Category	May Revenue	June Revenue	Growth	Flag
Beauty & Personal Care	235,710.83	238,323.37	1.11%	Not Flagged
Ethnic Wear	493,572.34	668,716.09	35.48%	Flagged
Home & Kitchen	494,285.21	479,068.32	-3.08%	Not Flagged
Kids Wear	243,977.89	186,472.78	-23.57%	Not Flagged
Western Wear	429,203.55	382,377.37	-10.91%	Not Flagged

9. Part 3 — Reliable AI Narrative and Data Masking

Part 3 communicates verified numbers safely using Context, Insight, and Implication. The narrative rules prohibit inventing causes, trends, sustainability, missing values, or unsupported business facts. Where information is insufficient, further investigation is explicitly required.

Reseller identifiers are masked into identifiers such as Reseller_001, Reseller_002, and Reseller_003. The masked view excludes raw reseller names. Because a private mapping table can reconnect the masked identifier to the original reseller, this is pseudonymization rather than complete anonymization.

10. Part 4 — Reliable Revenue Analysis Agent

Part 4 implements a local/mock revenue analysis agent that receives verified information from the Growth Engine and produces Context, Insight, Implication, and validation information. It does not connect to an external AI service and therefore requires no API key.

11. Agent Reliability Validation

```
AGENT RELIABILITY VALIDATION
=====
Output file exists: True
May categories verified: True
June category verified: True
Growth percentages verified: True

Agent validation PASSED.
```

12. Negative Testing

A negative test was performed by temporarily changing the required Growth Engine output filename. The agent then reported that the required input file was missing and stopped safely. The original filename was restored and the normal functional test passed.

13. Key Verified Findings

May 2026 categories exceeding the 8% threshold: Beauty & Personal Care (13.64%), Ethnic Wear (17.00%), and Western Wear (9.63%). June 2026: Ethnic Wear (35.48%).

Other June movements were Beauty & Personal Care 1.11%, Home & Kitchen -3.08%, Kids Wear -23.57%, and Western Wear -10.91%. A threshold flag indicates that the defined numerical condition was met; it does not establish the business cause of the movement.

14. Data Privacy and Reliability

- Data validation checks source integrity.
- Growth calculations are verified against expected results.
- Required input and output files are validated.
- Reseller identifiers are masked before narrative processing.
- Negative testing verifies missing-input behavior.
- The agent stops safely rather than generating unsupported results.
- Narrative rules separate verified facts from assumptions.

15. Reliability Principles

- Use verified source data.
- Preserve numerical accuracy.
- Separate facts from assumptions.
- Do not invent missing information.
- Validate calculations.
- Validate input and output files.
- Protect reseller identifiers through masking.
- Stop safely when required information is unavailable.
- Clearly identify limitations.
- Maintain reproducibility through deterministic data generation.

16. Limitations

- The dataset is simulated and does not represent production Meesho data.
- The analysis covers only April–June 2026.
- The dataset does not include every business variable, such as marketing spend, inventory, supplier information, customer acquisition cost, or product-level profitability.
- The current agent is a local/mock implementation rather than a production AI service.
- The pipeline does not connect to a live production database.

- Revenue movements are identified, but their causes are not independently established.
- The masking mechanism is pseudonymization rather than complete anonymization.

17. Future Enhancements

- Production database integration.
- Automated data ingestion and scheduled pipelines.
- Interactive dashboards and real-time KPI monitoring.
- Automated business alerts.
- Production AI model integration.
- Advanced anomaly detection.
- Additional business KPIs.
- Role-based access control and stronger privacy controls.
- Automated PDF and presentation generation.
- Historical trend and product-level profitability analysis.
- Inventory and availability analysis.

18. Conclusion

The Meesho Reseller Revenue Analytics project demonstrates how transactional data can be transformed into structured and validated business insights. The workflow combines SQL analysis, Python processing, Month-on-Month growth analysis, threshold-based flagging, data masking, evidence-based narrative generation, agent validation, negative testing, and safe failure handling.

The project emphasizes reliability by keeping generated narratives connected to verified numbers and by explicitly separating facts from assumptions. The current implementation provides a foundation that can be extended into a production analytics and AI-assisted reporting system.

19. Appendices

Appendix A — Project Structure

Capstone Project_Messo Reseller

```
■■■ .vscode
■■■ data
■■■ tasks
■■■ Part1_SQL
■■■ Part2_Engine
■■■ Part3_Narrative
■■■ Part4_Agent
■■■ Final_Project
■   ■■■ Report
■   ■■■ Presentation
■   ■■■ Evidence
■■■ meesho.db
■■■ README.md
```

Appendix B — Execution Commands

```
cd Part1_SQL
python main.py
```

Open Part1_SQL\Queries.sql using the SQLite extension in VS Code.

```
cd Part2_Engine
python Growth_Engine.py
```

```
cd Part3_Narrative
python Quick_Mask_Check.py
```

```
cd Part4_Agent
python Mock_Agent_Runner.py
```

Appendix C — Validation Evidence

The project includes validation for duplicate Order IDs, revenue calculations, MoM calculations, threshold flagging, masked output, agent output existence, expected categories, growth percentages, missing-input negative testing, and safe failure behavior.

Appendix D — Zero API Key Architecture

The complete current workflow runs locally through Python, SQLite, SQL, data masking, and the local/mock revenue analysis agent. No external API credentials are required.